

Victoria Xu

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Overview

Structural biology postdoctoral research associate at Imperial College London investigating virus DNA packaging mechanism using cryoEM.

Employment

Postdoctoral Research Associate

Imperial College London

London, UK

May 2026 - Present

Sarah L. Rouse lab

- Received funding from UKRI Impact Acceleration Account to work on improving AAV-vectors based on work produced during the PhD.

Research Assistant

Academia Sinica

Taipei, Taiwan

Jul 2020 - Nov 2020

Cheng-Hsun Ho lab

Education

PhD in Life Sciences

Imperial College London

Oct 2022 - May 2026

Supervisors: Sarah L. Rouse, Doryen Bubeck and Matthew Wake.

Examiners: Christian Speck (Imperial), Jonathan Grimes (Oxford).

Funding: UKRI-BBSRC and AstraZeneca.

- Structural biology research carried out in pharmaceutical and academic labs as well as national cryo-EM facility (eBIC).

MRes in Molecular and Cellular Biosciences

Imperial College London

Distinction

Oct 2021 - Oct 2022

Supervisors: Sarah L. Rouse and Doryen Bubeck

- Two 5-month-long research projects: 1. Studying mitochondrial inner-membrane fusion using cryoEM SPA and cgMD simulations. 2. Studying cellular defence against complement proteins using confocal microscopy.

BSc in Biochemistry

Imperial College London

2.1 Honors

Oct 2018 - Oct 2021

Publications and Grants

- Xu, V., McInnes, A., Wake, M., Acebrón-García-de-Eulate, M., Barritt, J. D., Bubeck, D., Rouse, S. L. (2026). Structural basis for Rep-mediated adeno-associated virus packaging. *Cell Reports*, 45(3), 117044. <https://doi.org/10.1016/j.celrep.2026.117044>
- Named Investigator on UKRI Impact Acceleration Account (IAA) grant proposal.

Conferences

Talks:

- Flatiron institute Structural and Molecular Biophysics (SMBp) seminar speaker. New York, NY, USA. 2025 (<https://indico.flatironinstitute.org/event/4067/>)
- Biophysical Society (BPS) annual meeting platform speaker. Los Angeles, CA, USA. 2025 ([https://www.cell.com/biophysj/abstract/S0006-3495\(24\)00992-5](https://www.cell.com/biophysj/abstract/S0006-3495(24)00992-5))
- Imperial Centre for Structural Biology open day speaker. London, UK. 2025

Session chair:

- BPS annual meeting protein-nucleic acid platform I. Los Angeles, CA, USA. 2025

Posters:

- The Collaborative Computational Project for Electron cryo-microscopy (CCP-EM) Spring symposium. Nottingham, UK. 2023-2026.
- American Society for Gene and Cell Therapy (ASGCT). New Orleans, LA, USA. 2025 (<https://www.sciencedirect.com/science/article/pii/S1525001625003648>)
- British Crystallographic Association winter meeting. Oxford, UK. 2023

Awards

- Eurofins Foundation Postgraduate Prize for MRes (Top student).
- Imperial Department of Life Sciences Travel Award to visit Flatiron institute.
- Bursary to attend 2026 CCPEM spring symposium.

Technical Experience

- **Wet lab:** Microscope operation (Glacios and Krios), grid-freezing (Vitrobot, Leica GP2, Chameleon high-speed grid vitrification robot), cloning, protein expression and purification (E.coli, mammalian cells, yeast), x-ray crystallography, mass photometry, confocal microscopy, ELISA, and virus production.
- **Dry lab:** Coarse-grained molecular dynamics simulation (CHARMM-GUI, GROMACS); SPA image processing (cryoSPARC and RELION), model building (ISOLDE, Coot and phoenix), deposition (PDB, EMDB and EMPIAR), Linux command line.
- **Teaching:** Supervised Structural Biology MRes students over 10-month project and Biochemistry undergraduate over 3-month project.

Referees

- Sarah L. Rouse: s.rouse@imperial.ac.uk
- Doryen Bubeck: d.bubeck@imperial.ac.uk
- Christian Speck: chris.speck@imperial.ac.uk